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30 SEPTEMBER 1966
SUPERSEDING
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19 JANUARY 1959

MILITARY SPECIFICATION

INITIATOR, CARTRIDGE ACTUATED, DELAY, M6A1, ASSEMBLY

1. SCOPE

1.1 This specification covers Initiator, Cartridge Actuated, Delay, M6A1, Assembly, loaded with Cartridge, Initiator, Delay, M46, for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATION

MILITARY

MIL-A-2550 — Ammunition and Special Weapons, General Specification For

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-109 — Quality Assurance Terms and Definitions

MIL-STD-1167 — Ammunition Data Card

MS-28741 — Hose, Assembly, Detachable End Fitting, Medium Pressure

DRAWINGS

MUNITIONS COMMAND

F8593316 — Initiator, Delay, M6A1, Assembly

C8796826 — Container, Ammunition, Fiber, M375 for Initiator, Delay, M4, M6A1, M12, M26, M33, M44, M45 or M49

C8796827 — Box, Packing, Ammunition, for Initiator, Delay, M4, M6A1, M12, M26, M33, M44, M45 or M49, in Fiber Container, M375

IEL-8593316 — Inspection Equipment List for Initiator, Delay, M6A1, Assembly

FSC 1377

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(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Materials, parts and assemblies shall comply with the requirements on the applicable drawings, referenced specifications and the following:

3.2 Breakaway torque. The chamber and cap of the assembled initiator shall withstand a breakaway torque application of 400 inch pounds without loosening or deformation.

3.3 Residual magnetism. The initiator shall not deflect the indicator of an approved compass more than three degrees in either direction when passed in front of the compass at a distance of six inches.

3.4 Functional.

3.4.1 *Ballistics.* Initiators loaded with accepted cartridges and temperature conditioned shall produce pressure values specified in table I. Pressure values shall be recorded at the end of three feet of size No. 4 aircraft hose in accordance with MS–28741.

TABLE I

Temperature	*Pressure (psi)*	*Ignition Delay (see 6.2)*
−65° to −70° F	1300 to 2200	2.50 seconds, max
65° to 75° F	1300 to 2200	1.80 to 2.20 seconds
160° to 165° F	1300 to 2200	1.60 seconds, min

3.4.1.1 *Initiation.* The initiator shall be fired by a source delivering an output equal to that of an M3A1 Initiator at the end of 15 feet of aircraft hose, size No. 4, standard MS–28741, at 70° ± 5°F.

3.4.2 The initiator shall not fail to fire.

3.4.3 The initiator shall function without mechanical failure (see 6.1).

3.5 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.5.1 *Metal defects.* The part shall be free from cracks, splits, cold shuts, inclusions, porosity or similar defects.

3.5.2 *Burr.* The part shall not have a burr which might interfere with the assembly or function of the item, or which might be injurious to personnel handling the item.

3.5.3 *Foreign matter.* The part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Quality assurance terms and definitions.* Reference shall be made to MIL–STD–109 to define quality assurance terms used.

4.1.2 *Inspection.* Inspection shall be in accordance with MIL–A–2550.

4.2 Inspection provisions.

4.2.1 *Lot.*

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4.2.1.1 *Submission of product.* The product shall be submitted in accordance with MIL–STD–105.

4.2.1.2 *Size of lot.* A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with a single lot of cartridges.

4.2.2 *Examination.* One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plan and associated criteria in table II. All nonconforming material shall be rejected. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of table II. Four power (4X) magnification may be used for inspection of such defects as burr and foreign matter.

TABLE II

Lot Size	*Sample Size*	*Major*		*Minor*	
		Accept	*Reject*	*Accept*	*Reject*
1 - 300	100% examination	-	-	-	-
301 - 800	50	0	1	1	2
801 - 1800	90	1	2	2	3

4.2.2.1 *Cartridge, Initiator, Delay, M46, Dwg D8594075.*

	Defect	*Method of Inspection*
CRITICAL		
1.	Mixed amunition	Visual
2.	Case split or perforated	Visual
3.	O-ring missing	Visual
4.	Sealing disc missing or damaged	Visual
5.	Marking missing, incorrect or illegible	Visual
MAJOR		
101.	Cartridge discolored, dirty, oily, corroded or stained	Visual
102.	Case dented	Visual
103.	O-ring split, cut or cracked	Visual
MINOR		
201.	Case scratched, wrinkled or bulged	Visual
202.	Varnish missing or inadequate	Visual

4.2.2.2 *Assembly, Dwg F8593316, D8594075, D8593313, D8594856, D8593314, B8359536 and B8593298.*

	Defect	*Method of Inspection*
CRITICAL		
1.	Setscrew missing	Visual

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	Defect	*Method of Inspection*
MAJOR		
101.	Thread damaged	Visual
102.	Burr	Visual
103.	Foreign matter	Visual
MINOR		
	None defined	

4.2.2.3 *Assembly, Dwg F8593316, D8593314, B8359536 and B8953298.*

	Defect	*Method of Inspection*
CRITICAL		
1.	Any part missing	Visual
2.	Improper assembly	Visual
3.	Metering groove of cartridge retainer missing	Visual
4.	Elongated hole of cartridge retainer missing	Visual
5.	Cartridge loose	Visual/Manual
6.	Cartridge retainer loose	Visual/Manual
MAJOR		
	None defined.	
MINOR		
	None defined.	

4.2.2.4 *Assembly, Dwg F8593316*, less B8359536 and B8593298.

	Defect	*Method of Inspection*
CRITICAL		
1.	Location of cartridge retainer	Gage
2.	Shearpin not engaged in firing pin	Gage
3.	Foreign matter in gas port	Visual
MAJOR		
101.	Length from firing pin housing boss face to gas port face, max	Gage
MINOR		
	None defined.	

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4.2.2.5 *Assembly, Dwg F8593816.*

	Defect	*Method of Inspection*
CRITICAL		
1.	Marking missing, incorrect or illegible	Visual
MAJOR		
101.	Height of cap, max	Gage
102.	Cap setscrew missing, loose or above flush	Visual
MINOR		
201.	Total length, max	Measure
202.	Location of labels incorrect	Visual
203.	Shipping cap or plug missing	Visual
204.	Barrier material missing from setscrew	Visual
205.	Protective finish inadequate or defective	Visual

4.2.2.6 *Container, Fiber, Dwg C8796826.*

	Defect	*Method of Inspection*
CRITICAL		
	None defined.	
MAJOR		
	None defined.	
MINOR		
201.	Gap between body and cover, max	Manual
202.	Contents loose	Manual
203.	Metal ends loose or distorted	Manual
204.	Cut or damaged through all impregnated layers	Visual
205.	Marking missing, incorrect or illegible	Visual
206.	Tape incomplete or badly wrinkled	Visual
207.	Length of tear tab, min	Visual
208.	Color of tape incorrect	Visual

4.2.2.7 *Box, Packing, Dwg C8796827.*

	Defect	*Method of Inspection*
CRITICAL		
	None defined.	

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	Defect	*Method of Inspection*
MAJOR		
	None defined.	
MINOR		
201.	Contents loose	Manual
202.	Hardware missing, loose, broken or improperly engaged	Visual/Manual
203.	Strapping missing, loose, broken or improperly assembled	Visual/Manual
204.	Contents of box exposed	Visual
205.	Box damaged	Visual
206.	Marking misleading or unidentifiable	Visual
207.	Number of units incorrect	Visual
208.	Car seal missing, unsealed or improperly positioned	Visual

4.2.3 *Inspection testing.* The following tests shall be performed in accordance with 4.3:

4.2.3.1 *Breakaway torque.* Each chamber and cap shall be subjected to the breakaway torque. Failure of any initiator to comply with the requirement of 3.2 shall be cause for rejection of the initiator.

4.2.3.2 *Residual magnetism.* Each initiator shall be subjected to a test for excessive residual magnetism. Failure of any initiator to comply with the requirement of 3.3 shall be cause for rejection of the initiator.

4.2.3.3 *Functional.* The following tests shall be performed by the Government:

4.2.3.3.1 *Ballistics.* Ballistic tests shall be performed in the quantity specified in table III. Failure of any initiator to comply with the requirements of 3.4.1, 3.4.2 and 3.4.3 shall be cause for rejection of the lot.

4.2.4 *Inspection equipment.* The examination and tests shall be made using equipment listed in IEL–8593316.

4.3 Test methods and procedures.

4.3.1 *Breakaway torque.* A torque wrench, set at the torque value specified in 3.2, shall be used to determine the adequacy of the joint. The chamber shall be secured in an air vise and the torque shall be applied to the cap.

4.3.2 *Residual magnetism.* The initiator shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the initiator six inches from and at the same height as an approved compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading as shown in figure 1. The initiator shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90° in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90° rotation of the initiator.

4.3.3 *Functional.*

4.3.3.1 *Ballistics.* Samples shall be selected, divided into test groups and conditioned as follows:

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TABLE III

First Production Lot

Lot Size	*Number of Samples*	*Conditioning Temperature*		
		−65° to −70° F	*65° to 75° F*	*160° to 165° F*
1 – 800	15	5	5	5
801 – 1800	30	10	10	10
		All Subsequent Lots		
1 – 800	10	4	2	4
801 – 1800	20	8	4	8

Upon completion of the required temperature conditioning the sample shall be fired, using an M3A1 Initiator assembled to the test initiator with 15 feet of size No. 4 aircraft hose, as specified in 3.4.1. No more than 5 minutes shall elapse from the time the samples are removed from the conditioning chamber and fired. Samples shall be reconditioned if this time is exceeded. When reconditioning initiators at minus 65°F, all condensation shall be removed from the interior surfaces before returning the initiator to the conditioning chamber. Required pressure readings shall be taken at the end of the 3-foot length of size No. 4 aircraft hose.

4.3.3.2 *Temperature conditioning.* A standard M641 Initiator, loaded with a dummy M46 Cartridge, containing a temperature sensitive element, shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking. Packing and marking shall be in accordance with drawings C8796826 and C8796827.

5.2 Data cards. Data card information shall be as specified in MIL–STD–1167.

6. NOTES

6.1 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.2 Ignition delay is defined as the time that elapses between the firing of the pressure element and the beginning of the continuous pressure rise on the pressure time record.

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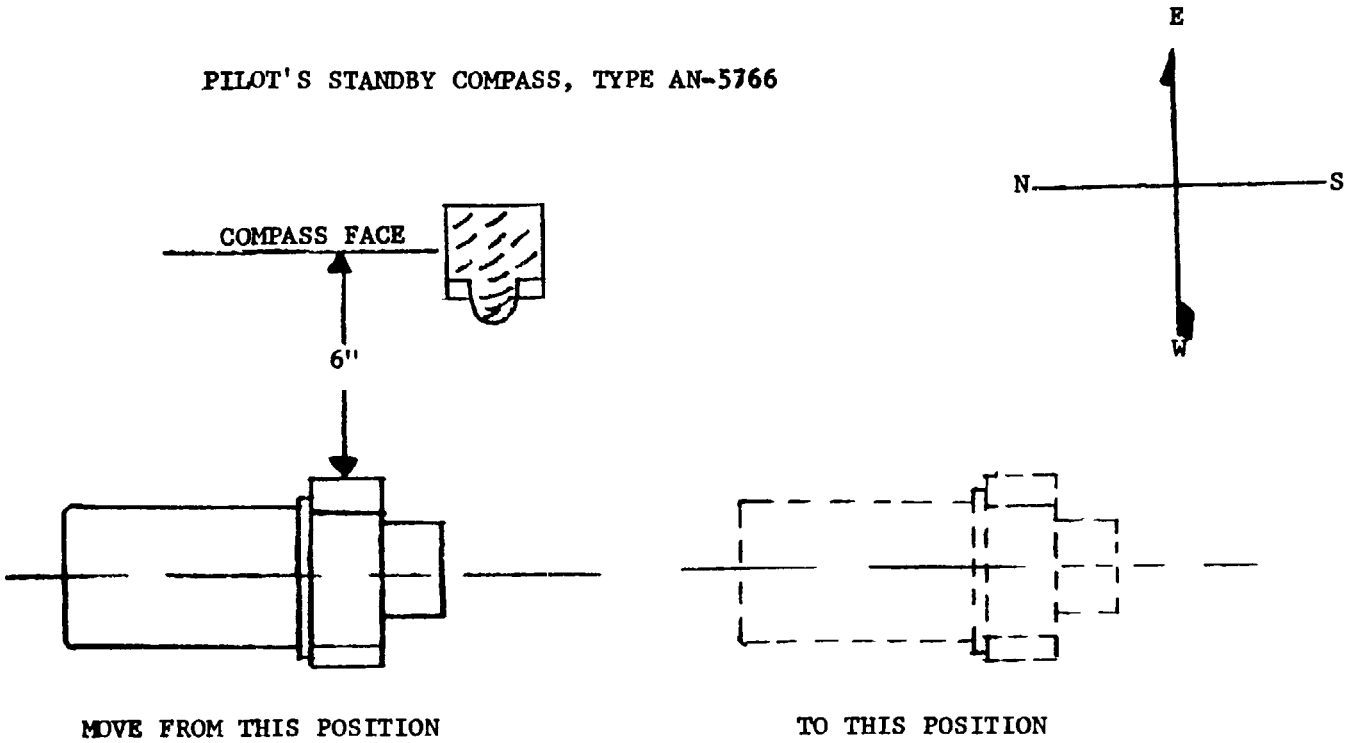
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PILOT'S STANDBY COMPASS, TYPE AN-5766



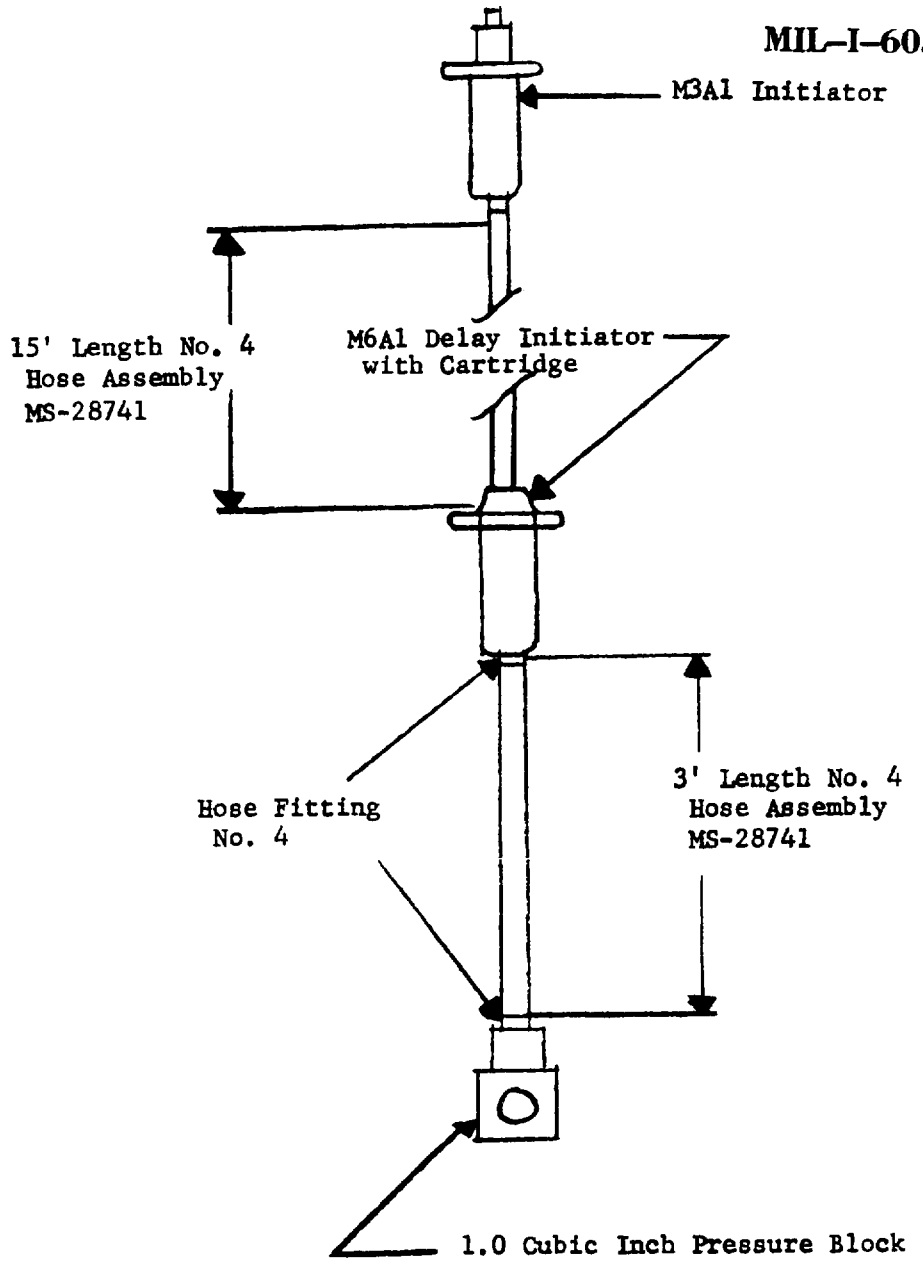
PLAN VIEW

NOTE:
INITIATOR AND COMPASS SHALL LIE IN A HORIZONTAL PLANE.

LAYOUT FOR DETERMINATION OF RESIDUAL MAGNETISM IN INITIATOR, DELAY, M6A1.

FIGURE 1

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PERFORMANCE TEST LAYOUT SKETCH

TOP VIEW

FIGURE 2

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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